WORKED SOLUTION

STEP III 1991, Question 2

Pure • Complex Numbers in Geometry

STEP III 1991, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q2

The key idea

A single complex number can carry both a ratio of lengths and an angle. The quotient $(z_1 - z_2)/(z_1 - z_3)$ records the ratio of the two sides of the triangle at P_1 (its modulus) and the turn from one side to the other (its argument), and those two pieces of data are exactly what the SAS test for similarity needs. Once the condition becomes a determinant, each part reduces to factorising a determinant, or to choosing a clever triangle $Q_1Q_2Q_3$ to compare with.

Solution

A misprint first

As printed, the left-hand denominator is $z_2 - z_3$. That can't be right: take $w_i = z_i$, so that the two triangles are identical. The right-hand side is then $(z_1 - z_2)/(z_1 - z_3)$, which differs from the left-hand side because $z_2 \neq z_1$. The intended condition, and the one that matches the determinant, is

$$\frac{z_1 - z_2}{z_1 - z_3} = \frac{w_1 - w_2}{w_1 - w_3}. \qquad (*)$$

This is what we prove.

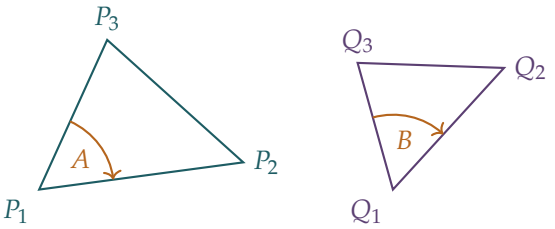
The similarity condition

Write A for the angle of triangle $P_1P_2P_3$ at P_1, so $0 < A < \pi$. In polar form,

$$\frac{z_1 - z_2}{z_1 - z_3} = \frac{z_2 - z_1}{z_3 - z_1} = \frac{P_1P_2}{P_1P_3}\,\mathrm{e}^{\mathrm{i}\theta},$$

where θ is the angle we turn through to get from the direction of $\overrightarrow{P_1P_3}$ to the direction of $\overrightarrow{P_1P_2}$. So $\theta = -A$ if $P_1 \to P_2 \to P_3$ runs anticlockwise (from P_1P_3 we turn clockwise to reach P_1P_2), and $\theta = +A$ if it runs clockwise. In short, the argument of this quotient

(taken in $(-\pi, \pi)$) has size equal to the angle at P_1 and sign that tells us the sense of the triangle. The same holds for the ws, with B the angle at Q_1.



Similar in the same sense $\Rightarrow (*)$. If the triangles are similar with $P_i \leftrightarrow Q_i$, the sides at P_1 and Q_1 are in the same ratio, $P_1P_2/P_1P_3 = Q_1Q_2/Q_1Q_3$, and the angles there agree, $A = B$. Because the two triangles run in the same sense, the two arguments $\mp A$ and $\mp B$ carry the same sign, so they are equal. Equal moduli and equal arguments give $(*)$.

$(*) \Rightarrow$ *similar in the same sense.* If $(*)$ holds, equating moduli gives $P_1P_2/P_1P_3 = Q_1Q_2/Q_1Q_3$. Equating arguments gives $A = B$, and also the same sign, so the triangles run in the same sense. Two sides in proportion with equal included angles is the SAS condition, so the triangles are similar with $P_i \leftrightarrow Q_i$. (If $(*)$ is real, both triples are collinear, and the degenerate "triangles" are again similar.)

Another way to see the same fact: $(*)$ says precisely that $w_i = az_i + b$ for the constants $a = (w_1 - w_2)/(z_1 - z_2) \neq 0$ and $b = w_1 - az_1$. The map $z \mapsto az + b$ is an enlargement by $|a|$ combined with a rotation by $\arg a$ and a translation, and such a map preserves shape and sense.

The determinant form

Clearing denominators (both are nonzero since the points are distinct), $(*)$ is equivalent to

$$(z_1 - z_2)(w_1 - w_3) - (z_1 - z_3)(w_1 - w_2) = 0.$$

Expanding, the z_1w_1 terms cancel and we are left with

$$z_1(w_2 - w_3) - z_2(w_1 - w_3) + z_3(w_1 - w_2) = 0,$$

which is the expansion of the determinant along its top row. So $(*)$ is the same as $\det = 0$.

For the parts it helps to evaluate the determinant another way. Subtract the first column from the second and third columns, which leaves the determinant unchanged, and then expand along the bottom row, which is now $(1, 0, 0)$:

$$\det \begin{pmatrix} z_1 & z_2 & z_3 \\ w_1 & w_2 & w_3 \\ 1 & 1 & 1 \end{pmatrix} = \det \begin{pmatrix} z_2 - z_1 & z_3 - z_1 \\ w_2 - w_1 & w_3 - w_1 \end{pmatrix}. \qquad (\dagger)$$

Part (i)

With $w_i = z_i^2$, (†) becomes

$$\begin{aligned}(z_2 - z_1)(z_3^2 - z_1^2) - (z_3 - z_1)(z_2^2 - z_1^2) &= (z_2 - z_1)(z_3 - z_1)[(z_3 + z_1) - (z_2 + z_1)] \\ &= (z_2 - z_1)(z_3 - z_1)(z_3 - z_2).\end{aligned}$$

Every factor is nonzero because the z_i are distinct. So the determinant is never zero, and the triangles are never similar (in the sense above).

Part (ii)

With $w_i = z_i^3$, use $z_j^3 - z_1^3 = (z_j - z_1)(z_j^2 + z_j z_1 + z_1^2)$:

$$\begin{aligned}(z_2 - z_1)(z_3^3 - z_1^3) - (z_3 - z_1)(z_2^3 - z_1^3) &= (z_2 - z_1)(z_3 - z_1)[(z_3^2 + z_3 z_1) - (z_2^2 + z_2 z_1)] \\ &= (z_2 - z_1)(z_3 - z_1)(z_3 - z_2)(z_1 + z_2 + z_3),\end{aligned}$$

since $z_3^2 - z_2^2 + z_1(z_3 - z_2) = (z_3 - z_2)(z_3 + z_2 + z_1)$. The first three factors are nonzero, so the determinant vanishes if and only if $z_1 + z_2 + z_3 = 0$. That is, the triangles are similar if and only if the centroid $\frac{1}{3}(z_1 + z_2 + z_3)$ is the origin.

Part (iii)

The trick is to compare the triangle with itself, relabelled. Take Q_1, Q_2, Q_3 to be P_2, P_3, P_1, that is

$$w_1 = z_2, \qquad w_2 = z_3, \qquad w_3 = z_1.$$

? **Ask yourself** Why should "equilateral" have anything to do with this relabelling?

Similarity with $P_1 \leftrightarrow P_2$, $P_2 \leftrightarrow P_3$, $P_3 \leftrightarrow P_1$ forces corresponding sides into a common ratio $k > 0$:

$$\frac{P_2P_3}{P_1P_2} = \frac{P_3P_1}{P_2P_3} = \frac{P_1P_2}{P_3P_1} = k.$$

Multiplying the three gives $k^3 = 1$, so $k = 1$ and all three sides are equal. Conversely, if the triangle is equilateral, the rotation through 120° about its centre (in the right direction) sends $P_1 \to P_2 \to P_3 \to P_1$. A rotation preserves shape and sense, so the triangle is similar to its relabelled self in the same sense. Hence

$$P_1P_2P_3 \text{ is equilateral} \iff \text{it is similar, in the same sense, to } P_2P_3P_1.$$

By the first part, the right-hand side holds if and only if

$$0 = \det \begin{pmatrix} z_1 & z_2 & z_3 \\ z_2 & z_3 & z_1 \\ 1 & 1 & 1 \end{pmatrix} = z_1(z_3 - z_1) - z_2(z_2 - z_1) + z_3(z_2 - z_3),$$

that is,

$$z_1z_3 - z_1^2 - z_2^2 + z_1z_2 + z_2z_3 - z_3^2 = 0 \iff z_2z_3 + z_3z_1 + z_1z_2 = z_1^2 + z_2^2 + z_3^2,$$

as required.

Checks

- Part (ii): $z_1, z_2, z_3 = 1, \omega, \omega^2$ (with $\omega = e^{2\pi i/3}$) have centroid 0, but their cubes all equal 1 and so aren't distinct. Try instead $z = 1,\ i,\ -1-i$. The centroid is 0 and the cubes are $1,\ -i,\ 2-2i$. Both triangles have sides $\sqrt{2}, \sqrt{5}, \sqrt{5}$ in the order P_1P_2, P_2P_3, P_3P_1 (here they are even congruent), and both run anticlockwise.
- Part (iii): for $z = 0, 1, e^{i\pi/3}$ both sides equal $e^{i\pi/3}$, since $1 + e^{2i\pi/3} = e^{i\pi/3}$.
- A computer test on 10^4 random triangles confirmed that $w_i = az_i + b$ always satisfies $(*)$ while the reflected triangle $w_i = a\bar{z}_i + b$ never does. It also confirmed both factorisations of $(\dagger)$ and the equilateral criterion. The condition as printed (with $z_2 - z_3$) failed in every one of the genuinely similar cases.

Common slips

- Ignoring the sense of the triangles. Moduli alone would only give similarity up to reflection. Mirror-image triangles have arguments of opposite sign, so they fail $(*)$.
- In part (iii), forgetting the converse: the relabelled-similarity argument has to run in both directions, and the step $k^3 = 1 \Rightarrow k = 1$ uses k being a positive real.

Beyond STEP

The maps $z \mapsto az + b$ are the orientation-preserving similarities of the plane. They are the simplest Möbius transformations, and the cross-ratio of four points generalises $(*)$. The equilateral condition also factorises as $(z_1 + \omega z_2 + \omega^2 z_3)(z_1 + \omega^2 z_2 + \omega z_3) = 0$, which says that z_1, z_2, z_3 run around an equilateral triangle in one sense or the other. Needham's *Visual Complex Analysis* develops this geometric way of thinking about complex numbers beautifully.

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